Exercise 1: Scope management (WBS) and schedule management (activity list)

Course	2110722 Software Project Management, semester 1/2569 (Assoc. Prof. Dr. Twittie Senivongse)
Exercise	Exercise 1 (5 points): Scope management (WBS) and schedule management (activity list)
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Project	Online shopping web site: Store front (Registration page, Product offering page, Checkout page) and Admin (Login, Order management)

Part 1: Work breakdown structure (WBS)

The WBS below is organised by **product** (deliverable-oriented, like the intranet WBS of Figure 5-3): level 1 is the project, level 2 is the set of major deliverables, and the five features given in the exercise appear as named WBS items under *1.4 Store Front* and *1.5 Admin*. Figure 1 is the chart form and Table 1 the tabular form. Shaded boxes in the chart and the rows marked **WP** in the table are the **work packages**, i.e. the items at the lowest level of the WBS.

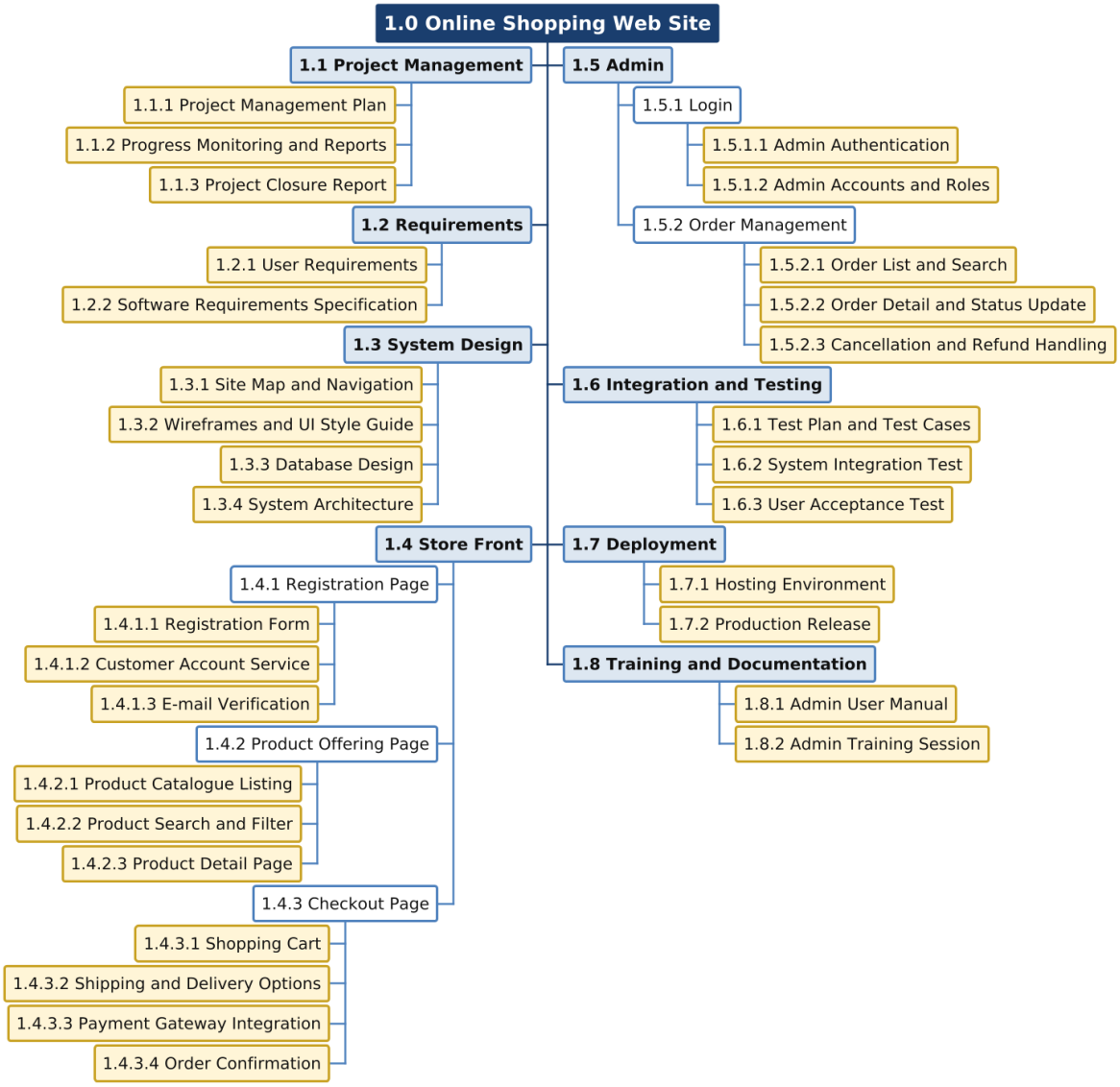


Figure 1: WBS of the online shopping web site project in chart form (organised by product; shaded leaf boxes are the work packages)

Table 1: WBS of the online shopping web site project in tabular form (WP = work package)

Level 1	Level 2	Level 3	Level 4	WP
1.0 Online Shopping Web Site				
	1.1 Project Management			
		1.1.1 Project Management Plan		WP
		1.1.2 Progress Monitoring and Reports		WP
		1.1.3 Project Closure Report		WP
	1.2 Requirements			
		1.2.1 User Requirements		WP
		1.2.2 Software Requirements Specification		WP
	1.3 System Design			
		1.3.1 Site Map and Navigation		WP
		1.3.2 Wireframes and UI Style Guide		WP
		1.3.3 Database Design		WP
		1.3.4 System Architecture		WP
	1.4 Store Front			
		1.4.1 Registration Page		
			1.4.1.1 Registration Form	WP
			1.4.1.2 Customer Account Service	WP
			1.4.1.3 E-mail Verification	WP
		1.4.2 Product Offering Page		
			1.4.2.1 Product Catalogue Listing	WP
			1.4.2.2 Product Search and Filter	WP
			1.4.2.3 Product Detail Page	WP
		1.4.3 Checkout Page		
			1.4.3.1 Shopping Cart	WP
			1.4.3.2 Shipping and Delivery Options	WP
			1.4.3.3 Payment Gateway Integration	WP
			1.4.3.4 Order Confirmation	WP
	1.5 Admin			
		1.5.1 Login		
			1.5.1.1 Admin Authentication	WP
			1.5.1.2 Admin Accounts and Roles	WP
		1.5.2 Order Management		
			1.5.2.1 Order List and Search	WP
			1.5.2.2 Order Detail and Status Update	WP
			1.5.2.3 Cancellation and Refund Handling	WP
	1.6 Integration and Testing			
		1.6.1 Test Plan and Test Cases		WP
		1.6.2 System Integration Test		WP
		1.6.3 User Acceptance Test		WP
	1.7 Deployment			

Level 1	Level 2	Level 3	Level 4	WP
		1.7.1 Hosting Environment		WP
		1.7.2 Production Release		WP
	1.8 Training and Documentation			
		1.8.1 Admin User Manual		WP
		1.8.2 Admin Training Session		WP

How the WBS was built. The WBS was created top-down (chapter 5, slide 23): starting from the whole web site, the scope was decomposed into the two product areas given in the exercise plus the supporting deliverables that any software project needs (project management, requirements, design, testing, deployment, training and documentation, as in the PMBOK example of Figure 5-4), and then each feature page was decomposed one more level into the parts that are actually built, for example *1.4.3 Checkout Page* into the shopping cart, the shipping step, the payment gateway integration and the order confirmation. Every item is named with a noun phrase as PMI recommends (slide 14); the verbs are reserved for the activities in Part 2. The 31 work packages are the lowest-level items, each unit of work appears in only one place, and the work content of each item is the sum of the items below it (slide 26), so the WBS defines the total scope of the project and can be numbered 1.0 / 1.1 / 1.1.1 exactly as in Table 5-4.

Part 2: Activity list for the work packages

Chapter 6 (slide 8) defines the activity list as a tabulation of the activities to be included on the project schedule, giving for each activity its **name**, its **identifier** and a **brief description**; the activities are driven by the deliverables on the WBS. Table 2 therefore lists, for every work package of Table 1, the activities needed to produce it. The activity identifier is the work package number followed by a running number (for example 1.4.1.1-A2 is the second activity of work package 1.4.1.1), so the schedule can always be traced back to the scope baseline. Activity names are verb phrases. The last column records one activity attribute, the **predecessor(s)**, using finish-to-start logic; a dash means the activity can start at the beginning of the project.

Table 2: Activity list of the online shopping web site project, by work package

1.1 Project Management, 1.2 Requirements and 1.3 System Design

Work package	ID	Activity	Brief description	Predecessor(s)
1.1.1 Project Management Plan	1.1.1-A1	Develop the project charter and scope statement	Agree the objectives, the five features in scope, constraints and assumptions with the shop owner.	–
	1.1.1-A2	Create the WBS, schedule and budget baseline	Decompose the scope into this WBS, estimate durations and cost, and baseline the plan.	1.1.1-A1
1.1.2 Progress Monitoring and Reports	1.1.2-A1	Monitor progress and report status	Weekly team meetings on status, issues and risks; status report to the shop owner every two weeks; control of change requests. Runs for the whole project.	1.1.1-A2
1.1.3 Project Closure Report	1.1.3-A1	Conduct the post-implementation review and close the project	Collect lessons learned, confirm that all deliverables are accepted and release the team.	1.7.2-A3
1.2.1 User Requirements	1.2.1-A1	Interview the shop owner and sample customers	Elicit which products are sold, the payment and delivery methods, and how orders are handled today.	1.1.1-A2

Work package	ID	Activity	Brief description	Predecessor(s)
	1.2.1-A2	Write user stories for the store front and admin	Capture registration, browsing, checkout and admin order-handling stories with acceptance criteria.	1.2.1-A1
1.2.2 Software Requirements Specification	1.2.2-A1	Specify the functional and non-functional requirements	Detail each of the five pages plus security, performance and browser-support requirements.	1.2.1-A2
	1.2.2-A2	Review the SRS with the shop owner and obtain sign-off	Walk through the SRS, resolve open points and baseline it (milestone M1).	1.2.2-A1
1.3.1 Site Map and Navigation	1.3.1-A1	Draw the site map and page flows	Store-front flow (register, browse, product, cart, checkout, confirmation) and admin flow (login, order list, order detail).	1.2.2-A2
1.3.2 Wireframes and UI Style Guide	1.3.2-A1	Sketch the wireframes and define the UI style guide	Low-fidelity layouts of the registration, product offering, checkout, admin login and order management pages; colours, typography, responsive grid and reusable components.	1.3.1-A1
1.3.3 Database Design	1.3.3-A1	Model the data entities and create the database schema	ER model of customer, product, category, cart, order, order item, payment and admin user; DDL scripts, indexes and seed data.	1.2.2-A2
1.3.4 System Architecture	1.3.4-A1	Select the technology stack, payment gateway and hosting	Choose the web framework, database, payment provider and cloud hosting, and document the architecture.	1.2.2-A2
	1.3.4-A2	Set up the source repository, CI pipeline and development environment	Repository and branching rules, automated build and test pipeline, shared development environment (milestone M2 once the design is approved).	1.3.4-A1

1.4 Store Front

Work package	ID	Activity	Brief description	Predecessor(s)
1.4.1.1 Registration Form	1.4.1.1-A1	Design the registration form	Detail the wireframe: fields (name, e-mail, password, phone, address), validation rules and error messages.	1.3.2-A1
	1.4.1.1-A2	Implement the registration form	Build the responsive page with client-side validation and a password-strength check.	1.4.1.1-A1, 1.3.4-A2
	1.4.1.1-A3	Unit-test the registration form	Test the validation of every field and the submission to the account service.	1.4.1.1-A2, 1.4.1.2-A1
1.4.1.2 Customer Account Service	1.4.1.2-A1	Implement the account creation service	Server-side API that stores the customer record with a hashed password and rejects duplicate e-mails.	1.3.3-A1, 1.3.4-A2
	1.4.1.2-A2	Implement customer sign-in and session handling	Sign-in and sign-out of registered customers, session cookie and password reset.	1.4.1.2-A1
	1.4.1.2-A3	Unit-test the account service	Test account creation, duplicate rejection, sign-in and session expiry.	1.4.1.2-A2
1.4.1.3 E-mail Verification	1.4.1.3-A1	Integrate the transactional e-mail service	Connect the e-mail provider and create the verification and notification templates.	1.3.4-A1
	1.4.1.3-A2	Implement the verification token and activation link	Generate a time-limited token on registration, e-mail it, and activate the account when the link is clicked.	1.4.1.2-A1, 1.4.1.3-A1
	1.4.1.3-A3	Test the registration flow end-to-end	Register, receive the e-mail, activate and sign in; test expired and reused tokens.	1.4.1.3-A2, 1.4.1.1-A3
1.4.2.1 Product Catalogue Listing	1.4.2.1-A1	Design the catalogue page	Product grid, category menu, sorting and pagination according to the style guide.	1.3.2-A1
	1.4.2.1-A2	Implement the product listing	Product and category tables, listing API and page with pagination and product images.	1.4.2.1-A1, 1.3.3-A1, 1.3.4-A2

Work package	ID	Activity	Brief description	Predecessor(s)
	1.4.2.1-A3	Load a sample catalogue and unit-test the listing	Import sample products with images; test the category and pagination logic.	1.4.2.1-A2
1.4.2.2 Product Search and Filter	1.4.2.2-A1	Implement keyword search and filters	Search by name and description; filter by category, price range and availability.	1.4.2.1-A2
	1.4.2.2-A2	Unit-test the search and filter results	Test ranking, empty results and combined filters.	1.4.2.2-A1
1.4.2.3 Product Detail Page	1.4.2.3-A1	Design the product detail page	Image gallery, description, price, stock status, quantity selector and add-to-cart button.	1.3.2-A1
	1.4.2.3-A2	Implement the product detail page	Page and API for a single product, stock display and the add-to-cart action.	1.4.2.3-A1, 1.4.2.1-A2
	1.4.2.3-A3	Unit-test the product detail page	Test the out-of-stock display, quantity limits and add-to-cart.	1.4.2.3-A2, 1.4.3.1-A1
1.4.3.1 Shopping Cart	1.4.3.1-A1	Implement the cart service	Add, change quantity and remove items; keep the cart per session and merge it into the account on sign-in.	1.4.2.1-A2, 1.3.3-A1
	1.4.3.1-A2	Implement the cart review page	Line items, subtotal, VAT and the proceed-to-checkout step according to the wireframe.	1.4.3.1-A1, 1.3.2-A1
	1.4.3.1-A3	Unit-test the cart calculations	Test totals, quantity limits against stock and the cart merge.	1.4.3.1-A2
1.4.3.2 Shipping and Delivery Options	1.4.3.2-A1	Implement the shipping address and delivery step	Address form pre-filled from the account, delivery methods and shipping-fee calculation.	1.4.3.1-A2, 1.4.1.2-A2
	1.4.3.2-A2	Unit-test address validation and shipping fees	Test required fields, postcode format and the fee of each delivery method.	1.4.3.2-A1
1.4.3.3 Payment Gateway Integration	1.4.3.3-A1	Set up the payment gateway sandbox	Merchant account, API keys and sandbox configuration of the chosen provider.	1.3.4-A1
	1.4.3.3-A2	Implement the payment request and result handling	Hosted payment page, callback verification, and handling of failure and timeout.	1.4.3.3-A1, 1.4.3.2-A1
	1.4.3.3-A3	Test payments with sandbox cards	Success, declined card, timeout and duplicate-callback cases.	1.4.3.3-A2
1.4.3.4 Order Confirmation	1.4.3.4-A1	Implement order creation after a successful payment	Create the order with items, address and payment reference, deduct stock and clear the cart.	1.4.3.3-A2
	1.4.3.4-A2	Implement the confirmation page and e-mail	Order number, summary and expected delivery shown on screen and sent to the customer.	1.4.3.4-A1, 1.4.1.3-A1
	1.4.3.4-A3	Unit-test order creation	Test stock deduction, order numbering and that a failed payment creates no order.	1.4.3.4-A2

1.5 Admin

Work package	ID	Activity	Brief description	Predecessor(s)
1.5.1.1 Admin Authentication	1.5.1.1-A1	Design the admin login page	Login form, error messages and forgotten-password link according to the style guide.	1.3.2-A1
	1.5.1.1-A2	Implement admin login	Password hashing, session timeout, lock-out after repeated failures and an audit log of logins.	1.5.1.1-A1, 1.3.3-A1, 1.3.4-A2
	1.5.1.1-A3	Unit-test admin login	Test valid and invalid credentials, lock-out and session expiry.	1.5.1.1-A2
1.5.1.2 Admin Accounts and Roles	1.5.1.2-A1	Implement admin account and role management	Create and deactivate admin users; roles Administrator and Order Staff with permission checks on every admin page.	1.5.1.1-A2

Work package	ID	Activity	Brief description	Predecessor(s)
	1.5.1.2-A2	Unit-test the role permissions	Test that Order Staff cannot manage accounts and that unauthenticated requests are rejected.	1.5.1.2-A1
1.5.2.1 Order List and Search	1.5.2.1-A1	Design the order management pages	Order list, filters and the order detail layout.	1.3.2-A1
	1.5.2.1-A2	Implement the order list	Paged list with search by order number, customer name or e-mail, and filter by status and date.	1.5.2.1-A1, 1.5.1.2-A1, 1.4.3.4-A1
	1.5.2.1-A3	Unit-test the order list and search	Test filters, search and access control.	1.5.2.1-A2
1.5.2.2 Order Detail and Status Update	1.5.2.2-A1	Implement the order detail and status workflow	Detail view; status transitions Paid, Packed, Shipped, Delivered with tracking number and staff name.	1.5.2.1-A2
	1.5.2.2-A2	Implement the status-change notification	E-mail the customer when the order is shipped or delivered.	1.5.2.2-A1, 1.4.1.3-A1
	1.5.2.2-A3	Unit-test the status transitions	Test allowed and forbidden transitions and the notification trigger.	1.5.2.2-A2
1.5.2.3 Cancellation and Refund Handling	1.5.2.3-A1	Implement order cancellation and refund	Cancel an unshipped order, request the refund through the payment gateway and restore stock.	1.5.2.2-A1, 1.4.3.3-A2
	1.5.2.3-A2	Unit-test the refund flow in the gateway sandbox	Test a full refund, a refund failure and stock restoration.	1.5.2.3-A1

1.6 Integration and Testing, 1.7 Deployment and 1.8 Training and Documentation

Work package	ID	Activity	Brief description	Predecessor(s)
1.6.1 Test Plan and Test Cases	1.6.1-A1	Write the test plan and system test cases	Derive system, performance and security test cases from the SRS; define entry and exit criteria.	1.2.2-A2
1.6.2 System Integration Test	1.6.2-A1	Integrate the store front and admin on the staging environment	Deploy all modules together with the gateway sandbox and the e-mail service (milestone M3, code complete).	1.4.1.3-A3, 1.4.2.2-A2, 1.4.2.3-A3, 1.4.3.4-A3, 1.5.1.2-A2, 1.5.2.3-A2, 1.7.1-A1
	1.6.2-A2	Execute the system test cases and log defects	Run the end-to-end scenarios (register, buy, admin ships, refund), the load test and the security checks.	1.6.2-A1, 1.6.1-A1
	1.6.2-A3	Fix defects and re-test	Correct defects by priority and run regression tests until the exit criteria are met.	1.6.2-A2
1.6.3 User Acceptance Test	1.6.3-A1	Conduct the UAT with the shop owner and staff	The owner and staff run the acceptance scenarios on staging with real product data.	1.6.2-A3
	1.6.3-A2	Obtain the acceptance sign-off	Record the UAT results and get written acceptance (milestone M4).	1.6.3-A1
1.7.1 Hosting Environment	1.7.1-A1	Provision the staging and production environments	Cloud servers, managed database, domain name and SSL certificate, plus monitoring, daily backup and security hardening.	1.3.4-A1
1.7.2 Production Release	1.7.2-A1	Load the production catalogue and admin accounts	Import the real product data and create the owner's and staff accounts in production.	1.6.3-A2, 1.7.1-A1
	1.7.2-A2	Deploy the release and go live	Deploy the accepted build, switch the payment gateway to live mode and verify a real test order (milestone M5).	1.7.2-A1

Work package	ID	Activity	Brief description	Predecessor(s)
	1.7.2-A3	Provide post-launch support	Two weeks of hypercare: monitor the site, fix urgent defects and hand over to maintenance.	1.7.2-A2
1.8.1 Admin User Manual	1.8.1-A1	Write the admin user manual	How to log in, process orders, update status, cancel and refund, and manage admin accounts.	1.5.2.3-A1
1.8.2 Admin Training Session	1.8.2-A1	Prepare the training materials and train the shop staff	Slides, exercises and demo orders on staging; half-day hands-on session on the admin system before go-live.	1.8.1-A1, 1.6.2-A1

Milestones

Milestones are significant events with no duration (chapter 6, slide 9); the five referenced in Table 2 are collected in Table 3 so that they can be put on the schedule and used to monitor progress.

Table 3: Milestone list

Milestone	Event	Reached when these activities finish
M1	Requirements approved (SRS signed off)	1.2.2-A2
M2	Design baseline approved	1.3.2-A1, 1.3.3-A1, 1.3.4-A2
M3	Code complete (all work packages of 1.4 and 1.5 unit-tested and integrated on staging)	1.6.2-A1
M4	Acceptance sign-off	1.6.3-A2
M5	Go-live	1.7.2-A2

How the activity list was derived. Each work package of Table 1 was decomposed into the specific actions that produce it (chapter 6, slide 8): for the feature work packages this is normally a design step, one or two implementation steps and a unit-test step, while the management and support work packages need only one or two activities. The 66 activities together cover all 31 work packages, and every activity belongs to exactly one work package, so the activity list is consistent with the scope baseline. The predecessor attribute records mandatory dependencies (for example a page cannot be tested before it is implemented, and the order list cannot be built before orders exist) as well as a few discretionary ones (for example page design follows the SRS sign-off); these dependencies are the input to the next step, sequencing the activities into a network diagram.